Problem 6 | Problems T6andT7_Electric and Magnetic Properties.pdf #6 | n-type silicon · conductivity

Phosphorus-Doped Silicon: Type and Conductivity

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 2

6. P is added to high purity silicon in order to introduce a concentration of 10^{+23} *m⁻³ of charge carriers at room temperature.*

- *a) What type of semiconductor is this material?*
- *b) What is the conductivity of this semiconductor at room temperature? Assume that the mobility of the e⁻ in the material is 0.14 m²/(V·s)*

Official Solution: b) 2243 Ω⁻¹m⁻¹

HYPOTHESES & PARAMETERS

Hypotheses

1. **P is a donor.** Phosphorus (group 15, 5 valence electrons) replaces silicon (4 valence electrons) and gives one electron to the conduction band, so the extrinsic silicon is n-type [Session 13 Slide 36].
2. **Electrons carry the current.** In n-type material $n_e \gg n_H$ and the conductivity is $\sigma \approx n_e|q_e|\mu_e$ [Session 13 Slide 36]. The "concentration of charge carriers" of the statement is the electron concentration n_e.
3. **Holes are negligible.** The holes of the intrinsic silicon are at most $n_i = 1.5 \times 10^{16}$ m⁻³ [Session 13 Slide 31], far fewer than the donor electrons; this is checked in step 5.
4. **Constant of the electron charge.** $|q_e| = 1.602 \times 10^{-19}$ C. Slide 10 prints 1.6×10^{-19} C; the key of the sheet (2243) is reproduced with 1.602×10^{-19} C.

Data

Quantity	Symbol	Value	In SI
Electron (carrier) concentration	n_e	10^{23} m⁻³	10^{23} m⁻³
Electron mobility	μ_e	0.14 m²/(V·s)	0.14 m²/(V·s)
Electron charge	$\lvert q_e \rvert$	1.602×10^{-19} C	same

Unknowns: (a) the type of semiconductor; (b) the conductivity σ.

MODEL & STRATEGY

Every donor atom adds one free electron, so the electron concentration equals the carrier concentration given by the statement. The current density follows from the drift velocity $v_d = \mu_e E$, and the conductivity is the proportionality constant between J and E (Ohm's law of the microscopic form, Slide 12). The figure shows the bar.

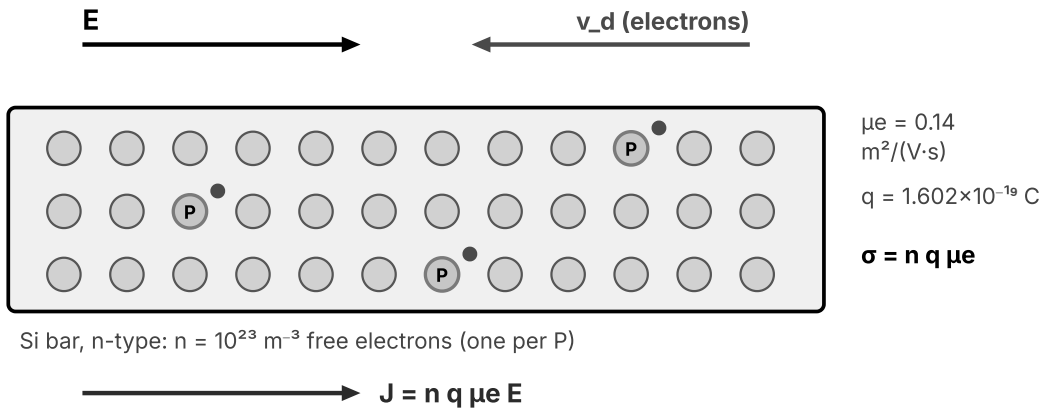


Figure 6. Problem 6: n-type silicon with 10^{23} m^{-3} conduction electrons from P donors; the field E drives the electrons against E, so the current density J points along E (schematic).

Strategy.

1. Classify the semiconductor from the group of P.
2. Write the current density from the drift velocity and read off the conductivity.
3. Evaluate the product in SI units, checking that the units reduce to $\Omega^{-1}m^{-1}$.
4. Check that the hole contribution is negligible.

STEP-BY-STEP CALCULATION

1. Type of the semiconductor (a) [Session 13 Slide 36]

Valence electrons of the dopant against the host:

$$Z_{\mathrm{P}} - Z_{\mathrm{Si}} = 5 - 4 = +1 \;\Rightarrow\; \text{donor} \;\Rightarrow\; \textbf{n-type extrinsic semiconductor}$$

2. Conductivity from the drift velocity [Session 13 Slides 11-12, 36]

Drift velocity and current density of n_e electrons of charge $|q_e|$, and Ohm's law $J = \sigma E$:

$$v_d = \mu_e \, E \qquad J = n_e \, |q_e| \, v_d = n_e \, |q_e| \, \mu_e \, E \qquad J = \sigma E$$

Comparing the last two expressions of the current density:

$$\sigma = n_e \, |q_e| \, \mu_e$$

3. Numerical evaluation (b)

Charge density of the free electrons, one operation at a time:

$$n_e \, |q_e| = 10^{23}\ \mathrm{m^{-3}} \times 1.602 \times 10^{-19}\ \mathrm{C} = 1.602 \times 10^{4}\ \frac{\mathrm{C}}{\mathrm{m^3}} = 16\,020\ \frac{\mathrm{C}}{\mathrm{m^3}}$$

Multiplication by the mobility:

$$\sigma = 16\,020\ \frac{\mathrm{C}}{\mathrm{m^3}} \times 0.14\ \frac{\mathrm{m^2}}{\mathrm{V\,s}} = 2242.8\ \frac{\mathrm{C}}{\mathrm{V\,s\,m}}$$

Reduction of the units, with $1\ \mathrm{C/s} = 1\ \mathrm{A}$ and $1\ \mathrm{V/A} = 1\ \Omega$:

$$\frac{\mathrm{C}}{\mathrm{V\,s\,m}} = \frac{\mathrm{A}}{\mathrm{V\,m}} = \frac{1}{\Omega\,\mathrm{m}} = \Omega^{-1}\mathrm{m}^{-1}$$

$$\boldsymbol{\sigma = 2243\ \Omega^{-1}\mathrm{m}^{-1}}$$

4. Resistivity (for comparison with the metals of Problem T6-01)

$$\rho = \frac{1}{\sigma} = \frac{1}{2242.8\ \Omega^{-1}\mathrm{m}^{-1}} = 4.459 \times 10^{-4}\ \Omega\,\mathrm{m}$$

5. The holes are negligible

Upper bound of the hole contribution, if there were n_i holes with the hole mobility of silicon (0.048 m²/(V·s) [Session 13 Slide 31]):

$$\sigma_h \leq p\,|q_e|\,\mu_h = 1.5 \times 10^{16}\ \mathrm{m^{-3}} \times 1.602 \times 10^{-19}\ \mathrm{C} \times 0.048\ \frac{\mathrm{m^2}}{\mathrm{V\,s}} = 1.153 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}$$

$$\frac{\sigma_h}{\sigma} \leq \frac{1.153 \times 10^{-4}}{2242.8} = 5.1 \times 10^{-8}$$

6. Effect of the rounded charge

With the value of Slide 10:

$$\sigma = 10^{23} \times 1.6 \times 10^{-19} \times 0.14\ \Omega^{-1}\mathrm{m}^{-1} = 2240\ \Omega^{-1}\mathrm{m}^{-1}$$

RESULT & CHECKS

Checks

- **Units.** $[n_e|q_e|\mu_e] = \mathrm{m^{-3}} \cdot \mathrm{C} \cdot \mathrm{m^2V^{-1}s^{-1}} = \Omega^{-1}\mathrm{m}^{-1}$ (step 3).
- **Order of magnitude.** 2243 $\Omega^{-1}\mathrm{m}^{-1}$ lies inside the semiconductor range of Slide 5 (10^{-6} to 10^{4} $\Omega^{-1}\mathrm{m}^{-1}$). Against the intrinsic value of silicon at room temperature, 4×10^{-4} $\Omega^{-1}\mathrm{m}^{-1}$ (the value used in Problem 8 of the sheet and printed on Slide 33; Slide 21 prints 4.5×10^{-4} $\Omega^{-1}\mathrm{m}^{-1}$):

$$\frac{\sigma}{\sigma_i} = \frac{2242.8\ \Omega^{-1}\mathrm{m}^{-1}}{4 \times 10^{-4}\ \Omega^{-1}\mathrm{m}^{-1}} = 5.6 \times 10^{6}$$

- **Hole share.** Below 10^{-7} of the total.
- **Limit.** For $n_e \to n_i$ the formula would need both carriers and tend to the intrinsic value; for $n_e \to 0, \sigma \to 0$.

Interpretation

The donors are a tiny fraction of the silicon atoms. The atom density follows from the density of Slide 31, the molar mass of silicon (28.09 g/mol, periodic table, not on the slides) and Avogadro's number:

$$N_{\mathrm{Si}} = \frac{\rho_m N_A}{M} = \frac{2.33 \times 10^{6}\ \mathrm{g/m^3} \times 6.022 \times 10^{23}\ \mathrm{mol^{-1}}}{28.09\ \mathrm{g/mol}} = 4.995 \times 10^{28}\ \mathrm{m^{-3}}$$

and the fraction of donors is:

$$\frac{n_e}{N_{\mathrm{Si}}} = \frac{10^{23}\ \mathrm{m^{-3}}}{4.995 \times 10^{28}\ \mathrm{m^{-3}}} = 2.0 \times 10^{-6}$$

that is, two donors per million atoms, and yet they raise the conductivity by almost seven orders of magnitude. The mobility of 0.14 m²/(V·s) is the one of the electrons in this material; Slide 42 shows that it falls when the dopant concentration is higher.

Check against the official solution

Item	Official	Computed	Comment
a) Type	not printed (the key lists b only)	n-type extrinsic	P is a group-15 donor, Slide 36
b) Conductivity	2243 Ω⁻¹m⁻¹	2243 Ω⁻¹m⁻¹	agrees with $q = 1.602 \times 10^{-19}$ C; with the rounded 1.6×10^{-19} C of Slide 10 the result is 2240